

Name: _____

INTRODUCTORY ACTIVITY STUDENT GUIDE (TAXONOMY example)

LIST OF CHARACTERISTICS (BALLS)

Using the collection of balls (see picture below and view the video link provided for more information), your task is to use a SYSTEMATICS approach to list and describe characteristics/qualities of the balls that you could use in creating a dichotomous key. You should create this list first using the space below. On the next page, you will use these characteristics to create a dichotomous key. You will need at least 5 characteristics to accurately and completely key out all of the balls, but should try and list as many as possible before you attempt to create your dichotomous key on the next page.



[Video of the 9 balls \(click me\)](#)

<https://lsu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=5b5a3928-d874-4d0f-a58c-ad9000db07d1>

CHARACTERISTICS

1. EXAMPLE >>> **COLOR** of balls (white, red, yellow, green, orange)
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.
- 15.

DICHOTOMOUS KEY (BALLS)

Using the list of characteristics you listed on the previous page for the various balls, create a DICHOTOMOUS KEY. I created a blank table for you to fill in below. Note, you may NOT need to use all 10 options below, but you need to COMPLETELY key out/identify each of the balls using the key you created below.

DICHOTOMOUS KEY FOR THE BALLS			
CHARACTER/TRAIT	BALLS	GO TO?	IDENTITY
1A:			
1B:			
2A:			
2B:			
3A:			
3B:			
4A:			
4B:			
5A:			
5B:			
6A:			
6B:			
7A:			
7B:			
8A:			
8B:			
9A:			
9B:			



LIST OF CHARACTERISTICS (COINS)

Penny



1 cent
No ridges
Copper & zinc

Nickel



5 cents
No ridges
Copper & nickel

Dime



10 cents
ridges
Copper & nickel

Quarter



25 cents
ridges
Copper & nickel

Dollar



1 dollar
No ridges
Copper & brass

Dollar



1 dollar
ridges
Copper & nickel

Half dollar



50 cents
ridges
Copper & nickel

Dollar

1 dollar
ridges
Silver



DICHOTOMOUS KEY (COINS)

Using the same approach as you used in the previous ball activity, create a DICHOTOMOUS KEY. I created a blank table for you to fill in below. Note, you may NOT need to use all 10 options below, but you need to COMPLETELY key out/identify each of the balls using the key you created below. In this activity, I listed the actual characteristics of the coins (i.e. size, composition, denomination, etc.), but there may be OTHER characters that you thought of and can use below that may be important in creating a dichotomous key.

DICHOTOMOUS KEY FOR THE COINS			
CHARACTER	COINS	GO TO?	IDENTITY
1A:			
1B:			
2A:			
2B:			
3A:			
3B:			
4A:			
4B:			
5A:			
5B:			
6A:			
6B:			
7A:			
7B:			
8A:			
8B:			
9A:			
9B:			
10A:			
10B:			

